INDUSTRY PROJECT

(Web and Mobile Apps)

08-Feb-2025

TECHNOLOGY STACK

- Python, Fast API
- MySQL (and Easy to migrate to PostgreSQL)
- ReactJS
- React Native

PROJECT LIST

01 Training and Reporting System

02 xxxx

03 xxxx

04 xxxx

05 xxxx

06 xxxx

Objective

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To digitize, standardize, and validate the end-to-end training lifecycle—ensuring workforce readiness, regulatory compliance, and verifiable skill development through real-time monitoring and transparent reporting for management and clients.

Mobile App: Learning and Field Execution

- **Dual-Mode Authentication:** Secure access via Mobile/Employee ID with SMS-OTP, supporting both individual and **Supervisor-led shared device** sessions.
- **Offline Learning Hub:** Encrypted access to study materials and results in "dead zones," featuring **Smart Archive** logic to optimize device storage.
- **Interactive Assessments:** Multi-lingual Pre and Post-training evaluations with **Dual Digital Signatures** to verify field-test integrity.
- **Geo-Fenced Attendance:** Biometric or QR-based check-ins restricted to authorized campus perimeters to eliminate proxy attendance.

Web Dashboard: Centralized Administartion

- **Dynamic Command Center:** A visual calendar to schedule, modify, and track "Live" vs. "Draft" training sessions across all sites.
- **Automated Compliance Engine:** Management of mandatory and periodic training with proactive alerts for expiring certifications and deployment blocks.
- **Content Management (LMS):** Centralized repository for multimedia study materials with trainer-controlled "Timed Release" functionality.
- **Master Data & Role Control:** Management of Company, Site, and User hierarchies with full audit trails for every system change.

Summary of Modules

Classroom Training

1. Master Data Management
2. Use Roles and Access Control
3. Mobile App Deployment and Authentication
4. Training Calendar, Dynamic Scheduling of Trainings & Curriculum Planning
5. Smart Invite and Enrollment Workflow
6. Shared Devise and Supervisor-Led Evaluation
7. Mobile App (Trainee, Trainer, and Supervisor)
8. Web App (Trainee & Trainer)
9. Content and Learning Management (LMS)
10. Assessment & Effectiveness Measurement
11. Mandatory & Periodic Training Compliance
12. Multi-Lingual & Regional Support
13. AI-Powered Visual Verification
14. Offline Functionality & Storage Management
15. Reporting, Analytics & Stakeholder Feedback

Online Self-Learning (E-Learning) Module

1. Self-Paced Course Architecture
2. AI-Driven Integrity & Facial Monitoring
3. Self-Learning Assessment & Mastery
4. Commercial Branding & Deployment (Self-Learning)
5. Commercial Branding & Deployment (Self-Learning)

1. Master Data Management (System Foundation)

- 1.1. **Entity Hierarchy:** Manage multi-tier data structures: Company > Region > City > Location/Site > Campus.
- 1.2. **Company Master:** Centralized database to manage multiple client organizations and internal business units
- 1.3. **Campus Mapping:** Detailed profiles for each campus including geo-coordinates, specific safety protocols, and client contact points.
- 1.4. **Training Category Master:** Classification of training types (Induction, Refresher, Specialized, Field-based).
- 1.5. **Topic Repository:** A master list of all available training modules with descriptions and duration with "Soft-Delete" capabilities to retire topics without losing historical data.
- 1.6. **Department Mapping:** Mapping training modeule list as per department (Security, Housekeeping, Logistics).
- 1.7. **Asset & Facility Master:** Tracking physical training resources (projectors, kits, field gear) and venue availability.
- 1.8. **Operational/Holiday Calendars:** Integration of regional/national holidays and client-specific shift cycles to prevent scheduling conflicts.
- 1.9. **White-Labeling**: Full Commercial Preference support including Company Logo integration and custom Color Themes to match corporate branding.
- 1.10. **Scalable Architecture**: Designed to support a single campus or scale across thousands of global sites.
- 1.11. Soft-Delete Architecture: Ensuring data can be deactivated without losing historical reporting integrity.

2. User Roles & Access Control

- 2.1. **Super Admin Role**: Full system access, configuration, and global data visibility.
- 2.2. **Trainer Role**: Access to schedules, trainee lists, material release, and assessment tools.
- 2.3. **Trainee Role**: Mobile-first access to personal schedules, study materials, and results.
- 2.4. **Client Role**: View-only access to reports and compliance data for their specific campus.
- 2.5. **Site Coordinator Role**: Local management of facilities and attendance verification.
- 2.6. **Granular Permissions**: Ability to toggle "Read," "Write," "Edit," and "Delete" for every sub-module.
- 2.7. **Profile Management**: Self-service portal for users to update basic info and view certifications.
- 2.8. **Device Binding**: Restricting trainee logins to a single mobile device to prevent proxy attendance.
- 2.9. **Audit Logs**: Tracking which user made changes to master data or training schedules.
- 2.10. **Session Timeouts**: Automatic logout for web and mobile interfaces to ensure data security.

3. Mobile App Deployment & Authentication

- 3.1. **Dual Distribution:** App availability via public App Stores and secure internal company servers (APK).
- 3.2. **Identity Verification:** Secure onboarding using Employee ID and Mobile Number cross-referenced against HR Master Data.
- 3.3. **SMS-Based Activation:** Automated OTP-based account activation and password recovery via an integrated SMS gateway.
- 3.4. **App Version Enforcement:** Mandatory update prompts to ensure all field personnel use the latest security and feature sets.
- 3.5. **Upload Photo and Matching**: Mandatory upload photo by snapping mobile camera and matching with the photo from database. Check liveness of photo.

4. Training Calendar, Dynamic Scheduling & Curriculum Planning

- 4.1. **Visual Training Calendar**: Monthly, weekly, and daily views of all active and planned sessions.
- 4.2. **Slot Management**: Defining start and end times with buffer periods for classroom transitions.
- 4.3. **Topic Mapping**: Linking specific curriculum topics to specific dates as per the annual training plan.
- 4.4. **Live Curriculum Editing**: Ability to modify topics or change trainers up until the session start time.
- 4.5. **Trainer Allocation**: Assigning qualified trainers based on topic expertise and location proximity.
- 4.6. **Field vs. Classroom Toggle**: Distinct markers for theory sessions and physical field drills.
- 4.7. **Recurring** Sessions: Tool to set up daily or weekly refresher courses automatically.
- 4.8. **Conflict Detection**: System alerts if a trainer or site is double-booked.
- 4.9. **Waitlist Management**: Automatically managing overflow if a training session reaches capacity.
- 4.10. **Draft Mode**: Allowing admins to build schedules in "Draft" before publishing to trainees.

5. Smart Invite & Enrollment Workflow

- 5.1. **Company-Based Filtering**: Bulk selection of trainees belonging to a specific client company.
- 5.2. **Location-Based** Selection: Inviting users based on their current assigned site or campus.
- 5.3. **Automated Invites**: System-generated emails and app notifications upon session publication.
- 5.4. **RSVP Tracking**: Real-time monitoring of who has accepted, declined, or not responded.
- 5.5. **QR Code Invitations**: Unique codes for trainees to scan at the venue for instant check-in.
- 5.6. **Manual Enrollment**: Admin ability to "Force Enroll" critical personnel into mandatory sessions.
- 5.7. **Prerequisite** Checks: Automatically blocking enrollment if a user hasn't completed "Level 1" training.
- 5.8. **Calendar Integration**: Option for users to sync training invites with Outlook or Google Calendars.
- 5.9. **Reminder Alerts**: Push notifications sent 24 hours and 1 hour before the session starts.
- 5.10. **Site-Transfer** Handling: Automatically updating training invites if a trainee is moved to a new campus.

6. Shared Device & Supervisor-Led Evaluation

- 6.1. **Supervisor Proxy Login:** Secure access for supervisors via OTP/Auth to conduct evaluations for trainees without personal mobile devices.
- 6.2. **Handover Logic:** A "Locked Exam Mode" allowing a supervisor to hand their tablet/phone to a trainee for assessment while blocking other app areas.
- 6.3. **Sequential Batch Processing:** Efficiently conducting tests for 20–30 trainees one by one on a single authorized device.
- 6.4. **Verification Signatures:** Capturing dual digital signatures (Supervisor & Trainee) to validate the integrity of shared-device exams.
- 6.5. **Upload Photo**: At the end of exam upload the selfie of employee or both (employee and supervisor together)

7. Mobile App (Trainee, Trainer, and Supervisor)

- 7.1. **Offline Mode**: Ability to view downloaded materials and schedules without active internet.
- 7.2. **Biometric Attendance**: Integration with phone fingerprint or face ID for secure check-in.
- 7.3. **In-App Document Viewer**: Secure viewing of PDFs and PPTs without allowing external downloads.
- 7.4. **Push Notifications**: Real-time updates on schedule changes or new material releases.
- 7.5. **Mobile Assessments**: Interface for taking pre/post-tests directly on the smartphone.
- 7.6. **Geo-Fencing**: Ensuring attendance can only be marked if the trainee is within the campus perimeter. However, Admin has right to override.
- 7.7. **Digital Signature**: Capturing signatures of both trainer and trainee upon session completion.
- 7.8. **Flashcards/Quick Bytes**: Micro-learning modules for on-the-go revision.
- 7.9. **Profile Dashboard**: Visual progress bar showing training completion percentages.
- 7.10. **Helpdesk Integration**: Quick-tap button to report technical issues or training queries.

8. Web App (Trainer & Admin)

- 8.1. **Executive Summary:** High-level charts showing total trained vs. pending personnel.
- 8.2. **Real-time Monitoring:** Dashboard showing "In-Progress" training sessions across all sites.
- 8.3. **Bulk Data Upload:** Excel/CSV import tools for users, topics, and master data.
- 8.4. **Announcement Banner:** Ability to push global alerts to all web and mobile users.
- 8.5. **Resource Heatmap:** Identifying which sites or trainers are over/under-utilized.
- 8.6. **Configuration Panel:** Settings for password policies, language defaults, and API integrations.
- 8.7. **Custom Branding:** Ability to upload company logos and theme colors.
- 8.8. **Approval Workflows:** Multi-stage approval for curriculum changes or budget requests.
- 8.9. **Integration Hub:** Managing connections to existing HRMS or Payroll systems.
- 8.10. **Feedback Moderation:** Reviewing and archiving trainer and trainee feedback.

9. Content & Learning Management (LMS)

- 9.1. **Pre-Study Release:** Tool to send specific materials 48–72 hours before the session.
- 9.2. **Timed Release:** Trainer-controlled "Unlock" button to release materials during a live lecture.
- 9.3. **Multi-Media Support:** Capability to host video tutorials, audio clips, and interactive PDFs.
- 9.4. **Version Archiving:** Keeping old versions of training content for legal/compliance audits.
- 9.5. **Content Tagging:** Categorizing files by "Field Work," "Theory," or "Safety."
- 9.6. **Knowledge Base:** A searchable library of all released materials for future reference.
- 9.7. **Watermarking:** Automatically adding trainee ID/Name to documents to prevent leaks.
- 9.8. **Video Progress Tracking:** Ensuring trainees watch the full video before it's marked "Complete."
- 9.9. **Interactive Scenarios:** Branching logic content where trainees choose actions in a field simulation.
- 9.10. **Copyright Management:** Setting expiry dates on sensitive training documents.

10. Assessments & Effectiveness Measurement

- 10.1. **Question Bank:** A centralized pool of MCQs, True/False, and descriptive questions.
- 10.2. **Randomization:** Shuffling questions and options to prevent cheating during assessments.
- 10.3. **Pre-Assessment:** Baseline tests to gauge existing knowledge before training begins.
- 10.4. **Post-Assessment:** Comprehensive tests to measure the learning "delta" or improvement.
- 10.5. **Automatic Grading:** Instant results for objective questions with feedback on wrong answers.
- 10.6. **Passing Thresholds:** Configurable minimum scores required for certification.
- 10.7. **Retake Policy:** System-enforced limits on how many times a trainee can attempt a test.
- 10.8. **Field Practical Scoring:** Trainer interface to grade physical performance during drills.
- 10.9. **Timer-Based Exams:** Setting strict time limits for online assessments.
- 10.10. **Proctoring Basics:** Alerts if a trainee tries to switch apps or minimize the test window.

11. Mandatory & Periodic Training Compliance

- 11.1. **Mandatory Tagging:** Ability to flag specific training modules as "Compulsory" based on job role, site assignment, or client-specific contract requirements.
- 11.2. **Periodic Frequency Engine:** Configurable settings to define the "Validity Period" for a course (e.g., Fire Safety every 6 months, First Aid every 1 year).
- 11.3. **Automated Re-enrollment:** System logic that automatically triggers a new training invitation 30 days before a trainee's current certification expires.
- 11.4. **Compliance Status Dashboard:** Real-time visual tracking showing "Compliant" vs. "Non-Compliant" percentages for every site and campus.
- 11.5. **Expiry Notifications:** Multi-stage alerts (SMS/Push/Email) sent to the trainee and their supervisor at T-minus 30, 15, and 5 days before certificate expiry.
- 11.6. **Escalation Matrix:** Automated reporting to higher management or HR if mandatory training remains overdue beyond a defined grace period.
- 11.7. **Deployment Blocking Logic:** Integration with site-roster systems to flag or "block" personnel from being deployed if their mandatory safety training has expired.
- 11.8. **Refresher Course Mapping:** Linking "Refresher" content to "Initial" content, ensuring the system knows that completing the 1-hour refresher counts as renewing the 8-hour basic course.
- 11.9. **Audit-Ready History:** Maintaining a timestamped log of every time a periodic training was successfully completed over the employee's entire tenure.
- 11.10. **Grace Period Management:** Ability for admins to grant temporary "Extensions" for mandatory training in case of emergencies or site-wide operational constraints.

12. Multi-Lingual & Regional Support

- 12.1. **Core Language Engine**: Support for English as the primary system language.
- 12.2. **Hindi Localization**: Full interface and content translation for North Indian regions.
- 12.3. **Marathi Support**: Specialized localization for Maharashtra-based operations.
- 12.4. **Southern Languages**: Inclusion of Tamil, Telugu, and Kannada for regional site coverage.
- 12.5. **Eastern Dialects**: Support for Bengali and Odia where applicable.
- 12.6. **Toggle Switch**: Easy-to-use language selector on the mobile app home screen.
- 12.7. Localized Assessments: Ability to take tests in the trainee's preferred language.
- 12.8. **Voice-Over Support**: Audio instructions in local languages for less tech-savvy users.
- 12.9. **Unicode Compatibility**: Ensuring all Indian scripts render correctly across all devices.
- 12.10. **Regional Date/Time Formats**: Displaying schedules as per local Indian standards.

13. AI-Powered Visual Verification

- 13.1. **Random Snapshots:** Front-camera triggers at intervals during tests to verify the trainee's active presence.
- 13.2. **AI Face Matching:** Real-time comparison of snapshots against the **Master Employee Database** for identity validation.
- 13.3. **Group Verification:** Mandatory trainer upload of batch photos to confirm physical classroom attendance.
- 13.4. **Integrity Alerts:** Automated flagging of **"No-Match"** or **"Multiple-Face"** scenarios for admin intervention.
- 13.5. **Anti-Proxy Logic:** Prevents "identity swapping" by monitoring the user throughout the assessment lifecycle.
- 13.6. **Geo-Location Tagging:** Precise GPS coordinates captured with every photo to ensure training occurs on-site.
- 13.7. **Audit-Ready Evidence:** Stores all visual proof within the session history for client and management audits.
- 13.8. **Pre-Publishing Alerts:** Triggers a notification to the trainer to review identity or location discrepancies before final results are released.
- 13.9. **Dual Sign-off:** Requires supervisor verification of flagged photos before results are officially published.
- 13.10. **Batch Integrity:** Matches group photo headcounts against digital check-ins to prevent attendance inflation.

14. Offline Functionality & Storage Management

- 14.1. **Local Encryption Engine:** Securely store downloaded study materials and result history in an encrypted local database for offline viewing.
- 14.2. **30-Day Inactivity Tracker:** Background monitor that identifies downloaded training modules not opened or accessed for a continuous 30-day period.
- 14.3. **Auto-Archive Logic:** Automated system trigger to remove local files of inactive courses to reclaim device storage space while keeping the "record" of completion visible.
- 14.4. **Manual Archive Control:** A "Storage Manager" interface within the app allowing trainees to manually archive or delete specific large files (videos/manuals) themselves.
- 14.5. **Proactive Intimation Alerts:** Push notifications sent to the trainee at day 25 and day 28 of inactivity, warning them of the upcoming auto-archiving of specific modules.
- 14.6. **On-Demand Restore:** A "Cloud-Fetch" button allowing users to re-download archived content immediately after a fresh authentication/login.
- 14.7. **Storage Usage Visualization:** A simple "Storage Bar" in the app settings showing how much space training materials are consuming versus available device memory.
- 14.8. **Offline Result Sync:** Ability to view past performance and scorecards without an active data connection once they have been synced at least once.
- 14.9. **Selective Download:** Allow users to choose between "High Quality" and "Data Saver" versions of study materials to optimize storage from the start.
- 14.10. **Background Cleanup Service:** System-level task that clears temporary cache and orphaned files during periods of low device activity to maintain app speed.

15. Reporting, Analytics & Stakeholder Feedback

- 15.1. **Trainee Progress Report**: Individual report cards showing scores, attendance, and feedback.
- 15.2. **Trainer Performance Index**: Analytics based on trainee scores and post-training feedback.
- 15.3. **Campus Compliance Report**: A dedicated report for clients showing "Safety Readiness" of the site.
- 15.4. **Gap Analysis**: Identifying topics where trainees consistently score low (needs more focus).
- 15.5. **Trainer Feedback Form**: Mandatory survey for trainees to rate the trainer's effectiveness.
- 15.6. **Management Feedback**: Qualitative input from trainers on trainee discipline and aptitude.
- 15.7. **Automated PDF Generation**: One-click export of all reports for email sharing.
- 15.8. **Scheduled Reports**: Monthly compliance summaries sent automatically to client stakeholders.
- 15.9. **Data Visualization**: Pie charts and bar graphs for "Training Completion" vs. "Target."
- 15.10. **Attendance Audit Trail**: Detailed log of check-in/check-out times and geo-locations.
- 15.11 **Feedback**: Following the conclusion of each training session, it is mandatory for every participant to submit a comprehensive **Session Evaluation and Feedback Form** to finalize their training record.

Supervisor’s Checklist: Shared Device Training

This checklist is designed for the Supervisor (or Trainer) who manages the assessment process on a single shared device. Sincœ this device is the "Source of Truth" for multiple people, following these steps ensures data integrity and prevents technical hiccups in the field.

Phase 1: Pre-Training Readiness (The "Zero" Hour)

- [ ] **Power & Storage:** Ensure the device has at least **40% battery** and check the "Storage Manager" to ensure enough space for offline result buffering.
- [ ] **Sync Check:** Open the app while on Wi-Fi/4G/5G to ensure the latest **Trainee Master Data** and **Question Banks** are downloaded.
- [ ] **App Version:** Verify that the app is updated to the latest version to avoid "Session Expired" errors mid-training.
- [ ] **Language Prep:** Confirm that the required regional languages (e.g., Hindi, Marathi, Kannada) are pre-loaded for the specific batch.

Phase 2: The Handover Protocol (Integrity Check)

- [ ] **ID Verification:** Physically verify the Trainee’s **Employee ID card** before selecting their name in the app.
- [ ] **Environment Check:** Ensure the trainee is in a quiet, well-lit area of the campus to prevent "guessing" or distraction.
- [ ] **Secure Mode Activation:** Before handing over the device, confirm the screen is in **"Handover Mode"** (Navigation bar hidden).
- [ ] **Instructions:** Briefly explain the "Swipe" and "Submit" actions to the trainee, especially if they are first-time users.

Phase 3: The Verification (The "Return" Check)

- [ ] **Visual Completion:** Upon receiving the device back, ensure the "Test Submitted" green tick is visible.
- [ ] **Dual Signature:** Ensure the trainee has provided their digital signature; then, provide your own **Supervisor PIN/Biometric** to unlock the main dashboard.
- [ ] **Qualitative Feedback:** Immediately log any "Field Observations" (e.g., trainee discipline or practical skill level) while the session is fresh.
- [ ] **Immediate Logout/Lock:** Ensure the session for *Trainee A* is fully closed before searching for *Trainee B* to prevent data overwriting.

Phase 4: Post-Session & Sync (Reporting)

- [ ] **Data Sync:** If training was done in a "Dead Zone," move to a networked area and tap **"Sync All"** to push results to the Web Admin Panel.
- [ ] **Archive Check:** If the session is finished, archive the study material if the 30-day limit is approaching to keep the device fast.
- [ ] **Report Generation:** Check the "Local Results" tab to ensure all 20–30 trainees in the batch have a "Success" status.

Online Self-Learning (E-Learning) Module

16. Self-Paced Course Architecture

- 16.1. **Course Enrollment Catalog:** A "Shop-style" interface where trainees browse and self-enroll in available or recommended modules.
- 16.2. **Modular Learning Paths:** Logical grouping of topics (e.g., Level 1 Security ➔ Level 2 Security) that must be completed in sequence.
- 16.3. **Progress Persistence:** Automated "Resume" functionality allowing trainees to stop a session and start exactly where they left off.
- 16.4. **Content Prerequisites:** System-enforced locking of advanced modules until foundational courses/tests are cleared.
- 16.5. **Interactive Video Overlay:** Integrated mid-video pop-up questions to ensure active engagement before the video continues.
- 16.6. **Drip-Feed Content:** Scheduled release of modules (e.g., one chapter per day) to prevent "speed-running" through critical safety training.

17. AI-Driven Integrity & Facial Monitoring

- 17.1. **Initial Identity Enrollment:** Capture of a "Master Reference Photo" during the first self-learning login using the front camera.
- 17.2. **Random Face Verification:** AI-triggered snapshots at random intervals during the course to ensure the registered trainee is still behind the screen.
- 17.3. **Anti-Spoofing Detection:** AI identification of "photos of photos" or masks to prevent fraudulent attendance.
- 17.4. **Multi-Face Alert:** Instant pausing of the course if more than one person is detected in the camera frame.
- 17.5. **Attention Tracking:** Monitoring eye-contact/presence; the course pauses automatically if the trainee moves away from the camera.
- 17.6. **Identity Discrepancy Flag:** Automated alert to the Trainer/Admin if the AI confidence score for a face match falls below the threshold.

18. Self-Learning Assessment & Mastery

- 18.1. **Instant Remediation:** If a trainee answers a question wrong, the system automatically suggests the specific slide/video section for review.
- 18.2. **Adaptive Questioning:** AI-driven difficulty adjustment based on the trainee's previous correct or incorrect answers.
- 18.3. **Randomized Mastery Tests:** Final exams pulled dynamically from the **Question Bank** to ensure no two trainees receive the same test.
- 18.4. **Peer-Ranked Leaderboards:** Optional gamification showing top scorers within a specific site or company to encourage healthy competition.
- 18.5. **Certificate Auto-Generation:** Immediate issuance of a digital, branded certificate upon successful completion and passing score.

19. Commercial Branding & Deployment (Self-Learning)

- 19.1. **White-Label Learner Portal:** Customizable UI colors and company logos for the self-learning interface to maintain brand identity.
- 19.2. **Multi-Site Scalability:** Ability to push specific self-learning modules to thousands of sites simultaneously with one click.
- 19.3. **Bandwidth Optimization:** Low-data mode for self-learning videos to accommodate field staff with limited mobile data plans.
- 19.4. **App Store & APK Sync:** Deployment of self-learning updates through both Google Play Store and secure internal server links.

20. Commercial Branding & Deployment (Self-Learning)

- 20.1. **Learning Velocity Reports:** Analytics on how long trainees take to complete specific modules compared to the average.
- 20.2. **Drop-off Analysis:** Identifying specific slides or videos where trainees most frequently stop or fail to progress.
- 20.3. **Pre-Publishing Integrity Alert:** A "Review Required" status for trainers to inspect facial match discrepancies before the final "Passed" status is published.
- 20.4. **Compliance Expiry Monitor:** Automated tracking of "Self-Learning" validity, triggering re-enrollment when a refresher is due.

Thank You

MANISH & ANIL

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